Shreeyash Kayastha

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SUMMARY

Computational Mathematics undergraduate at Kathmandu University with hands-on experience in Python, SQL, data visualization, statistical analysis, and time-series modeling. Built projects involving financial data analysis, predictive modeling, SQL-based analysis, and interactive dashboards. Seeking a Data Analyst internship or entry-level opportunity.

SKILLS

Languages: Python, SQL, R, C++
Data Analysis: Pandas, NumPy, Data Cleaning, Data Preprocessing, Statistical Analysis
Visualization: Tableau, Power BI, Excel (Pivot Tables, VLOOKUP), Matplotlib
Machine Learning: Scikit-Learn, XGBoost, Random Forest, Time-Series Analysis
Databases: MySQL, PostgreSQL
Tools: Git, GitHub, Google Colab, Jupyter Notebook

PROJECTS

Sales Data Dashboard | *2026*

- Built an interactive Tableau dashboard to analyze sales, profitability, and customer segments.
- Used SQL to extract, filter, and clean data for KPI and business performance analysis.
- Created KPIs and visualizations to identify sales trends and communicate business performance.

NEPSE Index Forecasting using XGBoost | *2026*

- Engineered lagged log-returns, RSI, rolling volatility, and moving-average features from historical NEPSE index data.
- Applied expanding-window walk-forward validation and Optuna hyperparameter optimization for XGBoost.
- Achieved an RMSE of 0.01345 and R^2 of 0.2063 on the reported evaluation, and compared performance with ARIMA and Ridge Regression.

Transport Management System | *2025–2026*

- Contributed to a university group project for managing transport-related data and application functionality.
- Worked on data management and system functionality as part of the software development team.

S&P 500 Market Direction Prediction | *2025*

- Analyzed historical S&P 500 data and developed a Random Forest classifier to predict next-day market direction.
- Engineered features using rolling averages, price ratios, and trend indicators.
- Evaluated predictions using walk-forward backtesting and precision.

Splitwise Expense Management Application | *2024–2025*

- Developed a C++ application for recording shared expenses and calculating balances between users.
- Applied object-oriented programming principles and used Git/GitHub for version control.

EDUCATION

Kathmandu University | Bachelor of Science in Computational Mathematics | *Expected 2027*

References

Dr. Rabindra Kayastha
Head of Department, Department of Mathematics
Kathmandu University
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